CS S5 Register Number: …………………..

Toc H INSTITUTE OF SCIENCE & TECHNOLOGY TIST

B. Tech Degree Internal Assessment Test – I, August 2025

CST 307 Microprocessor & Microcontrollers

Answer Key

Time: 120 Minutes Maximum Marks: 50

PART A

1. Describe the features of 8085 microprocessor **3**

- Any 3 features each carries 1 mark

•The Intel 8085 is an 8-bit general-purpose microprocessor.
•It has an 8-bit data bus.
•It has a 16-bit address bus, which addresses up to 64KB.
•It also has a 16-bit program counter & a 16-bit stack pointer.
•There are six 8-bit registers which are arranged in pairs: BC, DE, HL.
•It requires a voltage supply of +5V to operate at 3.2MHZ clock frequency
•The Intel 8085 comes as a 40-pin IC package.

2. Explain the pipelined architecture of 8086. **3**

Explain BIU and EU and the use of prefetch Queue

BIU-1.5 marsk

EU-1.5 marks

3. Explain any three signals used in maximum mode configuration in 8086

- Explanation for signals with suitable example each carries 1 Mark

4. The contents of different registers are given below. Obtain the effective addresses for different addressing modes. **3**

Offset(displacement)=5000H, [AX]=1000H, [BX]=2000H.

i) MOV [AX], [BX]
ii) MOV AX,5000[BX]

(ii) Register indirect

MOV AX, [BX]

DS:BX ⇔ 1000H:2000H
10H*DS ⇔ 10000
[BX] ⇔ + 2000
12000H - Effective address

(iii) Register relative

MOV AX, 5000 [BX]

DS: [5000 + BX]
10H*DS ⇔ 10000

Offset ⇔ + 5000
[BX] ⇔ + 2000
17000H - Effective address

5 **Summarize the 8086 instructions used for transferring data between registers, memory, stack and IO devices.** **3**

1. MOV- Move.
2. PUSH: Push to stack
3. POP: Pop from stack
4. XCHG: Exchange
5. IN: input code
6. OUT: Output to the port
7. XLAT: Translate
8. LEA instruction

- Explanation of data transfer /copy instructions in 8086

Expalin any 6,--0.5 marks each

PART B

(Answer all questions,)

6. a **Explain the internal Architecture of 8086 using block diagram** **10**

- Architecture – 5 Marks
- Explanation of each reg organization and remaining blocks carries – 1 Marks
  Accumulator
  Segment reg
  General Purpose re
  Instruction reg
  Flag reg

6. b **Compare architectural and signal difference between 8086 and 8088 microprocessors.** **4**

8086	8088
8086 is a 16 bit microprocessor.	8088 is a 16 bit microprocessor.
It has 16 bit data bus.	It has 8 bit data bus.
It has 16 bit ALU.	It has 16 bit ALU.
8086 requires memory banking to transfer 16 bit data at a time.	8088 does not require memory banking as it has an 8 bit data bus.
8086 performs faster memory operations as it can transfer 16 bits in one cycle.	8088 performs slower memory operations as it can transfer only 8 bits in one cycle.
8086 supports pipeline architecture.	8088 supports pipeline architecture.
8086 has a 6 byte pre-fetch queue for pipelining.	8088 has a 4 byte pre-fetch queue for pipelining.

- Any 4 architectural differences each carries 1 Mark

7.a **Illustrate the following arithmetic instructions in 8086** **7**

(a) CMP

(b) MUL

(c) DIV

(d) RCR

(e) ROR

(explain with example ,a,b,c carries 1 mark,d,e carries 2 marks)

7.b Explain addressing modes in 8086 with example. 7

- 1.IMMEDIATE addressing mode
- 2.DIRECT addressing mode
- 3.REGISTER addressing mode
- 4.REGISTER INDIRECT addressing mode
- 5.INDEXED addressing mode
- 6.REGISTER RELATIVE addressing mode
- 7.BASED INDEXED addressing mode
- 8.RELATIVE BASED INDEXED addressing mode
- 9.INTRASEGMENT DIRECT addressing mode
- 10.INTRASEGMENT INDIRECT addressing mode
- 11.INTERSEGMENT DIRECT addressing mode
- 12.INTERSEGMENT INDIRECT addressing mode

Different 12 addressing modes explanation of any 7 addressing modes with suitable examples each carries 1 Marks

8 **Develop an Assembly Language program to find smallest number in an array**. 7

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ASSUME CS:CODE, DS:DATA

DATA SEGMENT
LIST DB 52H, 23H, 56H, 45H   ; Array of numbers
COUNT EQU 4                  ; Number of elements in the array
SMALLEST DB 01H DUP(?)       ; Variable to store the smallest element
DATA ENDS

CODE SEGMENT
START:
   MOV AX, DATA           ; Initialize data segment
   MOV DS, AX

   MOV SI, OFFSET LIST      ; Point SI to start of LIST
   MOV CL, COUNT            ; Load counter (4 elements)
   MOV AL, [SI]           ; Assume first element as smallest

AGAIN:
   CMP AL, [SI+1]         ; Compare AL with next element
   JBE NEXT               ; If AL <= [SI+1], keep AL(JBE → Jump if Below or Equal)
   MOV AL, [SI+1]         ; Else update AL with smaller value

NEXT:
   INC SI              ; Move to next element
   DEC CL              ; Decrease counter
   JNZ AGAIN           ; Repeat until CL = 0

   MOV SI, OFFSET SMALLEST  ; Store result in SMALLEST
   MOV [SI], AL

   MOV AH, 4CH           ; Exit program
   INT 21H
CODE ENDS
END START
```